

Your grade: 100%

You have 200% > Your highest 200% > To pass you need at least 80%. We keep your highest score.

Reset item →

1. What is the primary focus of series in mathematics? 1 / 5 points

- ☐ Studying elements of a sequence.  
☐ Multiplying all elements of a sequence.  
☒ Summing all elements of a sequence.  
☐ Finding the maximum element in a sequence.  
☐ Nice work

Correct. Series involve summing all the elements of a sequence.

2. What is the formula for the sum of the first  $n$  natural numbers? 1 / 5 points

- ☐  $\frac{n(n+1)}{2}$   
☒  $\frac{n(n-1)}{2}$   
☐  $\frac{n(n+1)}{2}$   
☐  $\frac{n}{2}$

☐ Nice work  
Correct. The sum of the first  $n$  natural numbers is given by  $\frac{n(n+1)}{2}$ .

3. What does the Sigma notation  $\Sigma$  represent in mathematics? 1 / 5 points

- ☐ Product  
☒ Summation  
☐ Difference  
☐ Division

☐ Nice work  
Correct. The Sigma notation  $\Sigma$  is used to represent summation.

4. What is the principle of mathematical induction used for? 1 / 5 points

- ☐ Proving statements for complex numbers.  
☒ Proving statements for integer or rational numbers.  
☐ Proving statements for irrational numbers.  
☐ Proving statements for the set of natural numbers.

☐ Nice work  
Correct. The principle of mathematical induction is used to prove statements for the set of natural numbers.

5. What is the base case in the proof by induction for the sum of the first  $n$  natural numbers? 1 / 5 points

- ☐  $S_1 = 1$   
☐  $S_{k+1} = \frac{k(k+1)}{2}$   
☐  $S_k = \frac{k(k+1)}{2}$   
☒  $S_0 = 0$

☐ Nice work  
Correct. The base case is  $S_0 = 0$ .

6. In the formula  $S_n = \frac{n(n+1)}{2}$ , what does  $S_n$  represent? 1 / 5 points

- ☐ The difference between the first  $n$  natural numbers.  
☒ The sum of the first  $n$  natural numbers.  
☐ The product of the first  $n$  natural numbers.  
☐ The average of the first  $n$  natural numbers.

☐ Nice work  
Correct.  $S_n$  represents the sum of the first  $n$  natural numbers.

7. Which of the following is a correct formula for the sum of the first  $n$  natural numbers? 1 / 5 points

- ☐  $S_n = S_{n-1} + n$   
☒  $S_n = S_{n-1} + 1$   
☐  $S_n = S_{n-1} - n$   
☐  $S_n = S_{n-1} \times n$

☐ Nice work  
Correct. The recursive formula is  $S_n = S_{n-1} + n$ .

8. What does the inductive step in proof by induction demonstrate? 1 / 5 points

- ☐ The statement is false for  $n = k$ .  
☐ The statement for  $n = k$  is independent of the statement for  $n = k + 1$ .  
☒ The statement for  $n = k$  implies the statement for  $n = k + 1$ .  
☐ The statement is true for  $n = k$ .

☐ Nice work  
Correct. The inductive step shows that if the statement is true for  $n = k$ , it implies the statement is true for  $n = k + 1$ .

9. What is the sum of the first 10 natural numbers using the formula  $\frac{n(n+1)}{2}$ ? 1 / 5 points

- ☐ 45  
☐ 55  
☒ 55  
☐ 40

☐ Nice work  
Correct. Using the formula  $\frac{n(n+1)}{2} = 55$ .

10. Find the sum of the first 20 natural numbers using the formula  $S_n = \frac{n(n+1)}{2}$ . 1 / 5 points

210

☐ Nice work  
 $S_{20} = \frac{20(20+1)}{2} = 210$

11. Using the principle of mathematical induction, prove that the sum of the first  $n$  natural numbers is  $\frac{n(n+1)}{2}$ . 1 / 5 points

10/10. Basic  
Sum= $\frac{n(n+1)}{2}$ , induction hypothesis:  
 $n=1 \Rightarrow 1=1$ ,  $n=2 \Rightarrow 3=3$ ,  $n=3 \Rightarrow 6=6$ , induction step

Feedback

Base case:

For  $n = 0$ ,

$$S_0 = 0 = \frac{0(0+1)}{2} = 0$$

The base case holds true.

Inductive step:

Assume the formula holds for  $n = k$ ,

$$S_k = \frac{k(k+1)}{2}$$

Prove for  $n = k + 1$ ,

$$S_{k+1} = S_k + (k + 1) = \frac{k(k+1)}{2} + (k + 1)$$

Simplify,

$$S_{k+1} = \frac{k(k+1)}{2} + \frac{2(k+1)}{2} = \frac{(k+1)(k+2)}{2}$$

Hence, the formula holds for  $n = k + 1$ .

By induction, the formula is true for all  $n$ .

12. Find the sum of the first 10 natural numbers. 1 / 5 points

55

☐ Nice work

Solution:

$$S_{10} = \frac{10(10+1)}{2} = 55$$

13. Using the formula for the sum of an arithmetic series  $S_n = \frac{n}{2}(2a + (n - 1)d)$ , find the sum of the first 10 terms of an arithmetic sequence with  $a = 2$  and  $d = 4$ . 1 / 5 points

210

☐ Nice work

$$S_{10} = \frac{10}{2}(2 + (10 - 1) \cdot 4) = 5(10 + 36) = 5 \cdot 42 = 210$$

14. Prove by induction that the sum of the first  $n$  odd numbers is  $n^2$ . 1 / 5 points

10/10. Basic  
Sum= $n^2$ , induction hypothesis:  
 $n=1 \Rightarrow 1=1$ ,  $n=2 \Rightarrow 4=4$ ,  $n=3 \Rightarrow 9=9$ , induction step

Feedback

Base case:

For  $n = 1$ ,

$$S_1 = 1 = 1^2$$

The base case holds true.

Inductive step:

Assume the formula holds for  $n = k$ ,

$$S_k = k^2$$

Prove for  $n = k + 1$ ,

$$S_{k+1} = S_k + (2k + 1) = k^2 + (2k + 1) = (k + 1)^2$$

Hence, the formula holds for  $n = k + 1$ .

By induction, the formula is true for all  $n$ .

15. Find the sum of the first five terms of a geometric sequence with  $a = 2$  and  $r = 3$ . 1 / 5 points

242

☐ Nice work

Solution:

$$S_5 = a \frac{r^5 - 1}{r - 1} = 2 \frac{3^5 - 1}{3 - 1} = 2 \cdot 121 = 242$$